

the neocortical column incorporates detailed representations of a minimum of 30,000 neurons. Generally, in order to facilitate valid boundary conditions, N times this number is required.

In early stages, IBM BlueGene/L supercomputer running on 8192 processors was being used. BlueGene/L system is a totally new approach in supercomputer design optimised for bandwidth, scalability and the ability to handle large amounts of data while consuming a fraction of power and floor space required by some of the leading supercomputing systems.

The system needs the floor space of about four large fridges, and has a peak processing speed of a minimum of 22.8 trillion floating-point operations per second (22.8 teraflops). This means that the supercomputer can theoretically carry out 22.8 trillion calculations per second. By mapping one or two simulated brain neurons to each processor, the computer becomes a silicon replica of 10,000 neurons communicating back and forth.

BlueGene/L System was designed to simulate high-speed atomic interactions, which also provide the optimal architecture to simulating neural interactions. Simulations optimised for clusters using MPI messaging can easily be ported to run on Blue Gene. BlueGene/L Prototype System allows parallel processing of virtually any number of processors to meet the memory and speed demands of a simulation. It can be scaled up enormously to meet further computational demand, and has provided the foundation for further development on BlueGene/P, the next-generation IBM supercomputer that constitutes a quantum leap in memory capacity, processing speed and whole-brain simulations.

Till recently, the project was powered by a 16,384-core IBM BlueGene/P supercomputer that



Fig. 6: 3D neuron morphology reconstruction in progress (Image courtesy: www.artificialbrains.com/blue-brain-project)

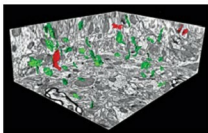


Fig. 7: Three-dimensional reconstruction of synapses. The colour shows whether the synapse is symmetric (red) or asymmetric (green) (Image courtesy: www.upm.es)

had a memory about eight times more than the memory of IBM BlueGene/L. This makes IBM BlueGene/P capable of touching pet flops speeds and quadrillions of calculations per second.

Presently, IBM BlueGene/Q supercomputer with 65,536 cores and extended memory capabilities hosted by Swiss National Supercomputing Center in Lugano has been deployed.

The Blue Brain Software

The human neocortex region has millions of microcircuits called neocortical columns and, hence, it is important to create a molecular-level modelling of a neocortical column using sophisticated software. This software version is transformed into a hardware version—a molecular-level neocortical column chip—which can then be duplicated. While there are many software for simple/point neurons, there are no optimised software programs that can carry

out very large-scale, that is, tens of thousands, simulations of morphologically-complex neurons. The software for such simulations consists of a hybrid between two powerful software approaches: one for large-scale neural network simulations called Neocortical Simulator and the other is a well-established program called NEURON.

Microcircuit databases are critical as Laboratory of Neural Micro Circuitry has attained a huge quantum of data on the

composition and connectivity of the neocortical column. Microcircuit data from many other labs all over the globe will also be included. The new database, NEOBASE, is constructed on ROOT platform and modelled on the lines of CERN's innovative work that has facilitated thousands of researchers to work as one team on the same project.

The ultimate objectives are to enable full-scale researcher interactions through NEOBASE for further construction of microcircuit database, partnered visualisations and planned simulations. Microcircuit visualisation is possible since BlueBuilder has been built to design, upload and connect thousands of model neurons. BlueBuilder uploads Neurolucida files with complete morphological data.

In parallel, the model neurons need to be modelled in NEURON to include the physiological properties. The export from BlueBuilder references NEURON files for BlueColumn simulations, while BlueBuilder pulls out neurons directly from NEOBASE. Connections are formed according to established connectivity rules. Phenomenological and biophysical models of synapses are assigned to the connections within BlueBuilder. It generates two types of output files: first for simulations by NCS or Neurodamus on Blue Gene and second for visualisation in BlueVision.

Visualisation consists of diverse